

Muhammad Hassan Umar

Dubai, UAE | UAE Golden Visa Holder
+971 50 232 6079 | meet.mhassanumar@gmail.com
linkedin.com/in/mhumar2568574 | usheh05.github.io/mhassanumar.github.io

Mechanical engineering student (91 WAM, Dean's List) with industry experience at Mercedes-Benz and in space-mechanism research. Seeking an internship or working-student role in mechanical design, manufacturing or energy systems.

EDUCATION

University of Wollongong Dubai

Bachelor of Engineering (Honours) in Mechanical Engineering

Dubai | UAE

Apr 2025 – Mar 2029

- 91 WAM (High Distinction Average); Dean's List Recipient 2025 (Top 5% of students).
- Relevant Coursework: Engineering Mechanics, Materials in Design, Engineering Design for Sustainability, Advanced Engineering Mathematics, Engineering Fluid Mechanics.

Karlsruhe Institute of Technology

Bachelor of Science in Mechanical Engineering

Karlsruhe | Germany

Oct 2023 – Mar 2025

- Relevant Coursework: Mechanical Design I & II, Materials Science, Global Production and Management, Basics of Manufacturing Technology.

PROFESSIONAL EXPERIENCE

Mercedes-Benz AG

Engineering Intern

Sindelfingen | Germany

Nov 2024 – Jan 2025

- Designed a 4-tier Power BI dashboard system enabling real-time production scrap visibility, projecting a 63% reduction in scrap costs in year 1 and 74-83% over 4-5 years.
- Applied moving averages and Pareto weighting to production defect data to identify and prioritize highest-impact scrap sources across production lines.
- Presented to a 3-member panel of Mercedes-Benz engineers and board members and placed first of 6 teams.

Aero Pulse System

Research Intern

Dubai | UAE

Jun 2026 – Present

- Authored a technical paper on Miura-ori and Kresling origami-inspired robotic arms for space, assessing how launch loads, thermal cycling and fatigue shape the design.
- Developing a framework matching six robotic capabilities to the CubeSat sizes that can support them by mass, volume and power.

ESCCO

BIM Assistant

Jeddah | KSA

Jul 2026 – Aug 2026

- Checked Revit models of Hyundai passenger and service lifts for Four Seasons hotel projects against 2D drawings, correcting elements across buildings with up to 7 lifts.
- Completed and verified equipment schedules for 4 drawing sets, covering around 30 parameters per component.

PROJECTS

Stellum: Sustainable Manufacturing Plant Design

Engineering Design for Sustainability

Dubai | UAE

Grade: 98

- Designed a six-zone manufacturing plant layout and a stainless-steel container in AutoCAD for full end-of-life recyclability, within a \$10M CAPEX.

MATLAB & Python

Engineering Computing and Analysis

Dubai | UAE

Grade: 100

- Built a MATLAB App Designer trajectory solver with 3D plots and collision detection, and hand-validated Python beam solvers for shear-force and bending-moment diagrams.

SKILLS AND PERSONAL

Technical Skills: Fusion 360 | AutoCAD | Revit | MATLAB | Python | NumPy | Pandas | Matplotlib | Power BI

Languages: English (C2), Urdu (Fluent), French (B2), Arabic (Conversational), German (A1)